

PROGRAM – 2

Develop a Multi-Stage Dockerfile for Container Orchestration

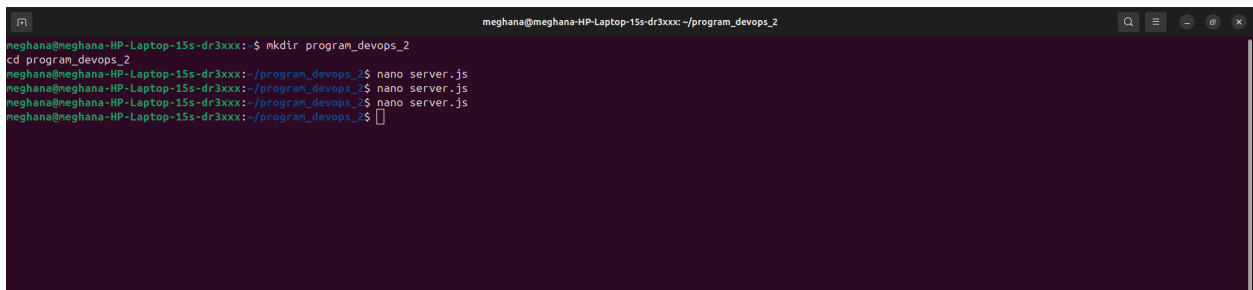
Project Name: `program_devops_2`

Technology Used: Node.js, Docker

Project Structure

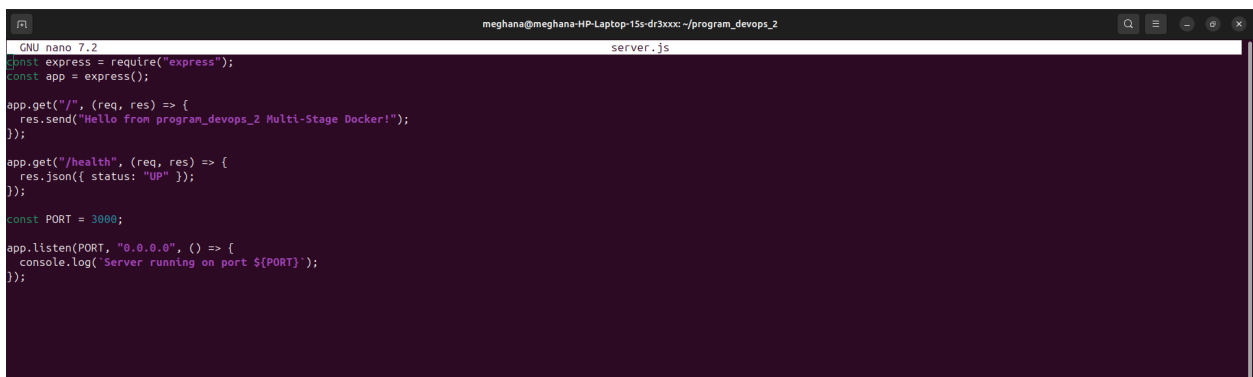
`program_devops_2/`

- |
- |— server.js
- |— package.json
- |— Dockerfile



```
meghana@meghana-HP-Laptop-15s-dr3xxx: ~/program_devops_2
meghana@meghana-HP-Laptop-15s-dr3xxx: $ mkdir program_devops_2
cd program_devops_2
meghana@meghana-HP-Laptop-15s-dr3xxx: ~/program_devops_2 $ nano server.js
meghana@meghana-HP-Laptop-15s-dr3xxx: ~/program_devops_2 $ nano server.js
meghana@meghana-HP-Laptop-15s-dr3xxx: ~/program_devops_2 $ nano server.js
meghana@meghana-HP-Laptop-15s-dr3xxx: ~/program_devops_2 $
```

Step 1 – Node.js Application (server.js)



```
GNU nano 7.2 server.js
const express = require("express");
const app = express();

app.get("/", (req, res) => {
  res.send("Hello from program_devops_2 Multi-Stage Docker!");
});

app.get("/health", (req, res) => {
  res.json({ status: "UP" });
});

const PORT = 3000;

app.listen(PORT, "0.0.0.0", () => {
  console.log(`Server running on port ${PORT}`);
});
```

Step 2 – Dependency File (package.json)

```
meghana@meghana-HP-Laptop-15s-dr3xxx: ~/program_devops_2
GNU nano 7.2 package.json
{
  "name": "program_devops_2",
  "version": "1.0.0",
  "main": "server.js",
  "scripts": {
    "start": "node server.js"
  },
  "dependencies": {
    "express": "^4.18.2"
  }
}
```

Step 3 – Multi-Stage Dockerfile

```
meghana@meghana-HP-Laptop-15s-dr3xxx: ~/program_devops_2
GNU nano 7.2 Dockerfile
# ----- Stage 1: Build -----
FROM node:18-alpine AS builder

WORKDIR /app

COPY package.json .
RUN npm install

COPY server.js .

# ----- Stage 2: Production -----
FROM node:18-alpine

WORKDIR /app

COPY --from=builder /app .

EXPOSE 3000

CMD ["npm", "start"]

[ Wrote 20 lines ]
~? Help      ~? Write Out  ~? Where Is   ~? Cut        ~? Execute    ~? Location   ~? Undo       ~? Set Mark   ~? To Bracket ~? Previous  ~? Back
~? Exit      ~? Read File  ~? Replace    ~? Paste       ~? Justify    ~? Go To Line ~? Redo       ~? Copy       ~? Where Was  ~? Next      ~? Forward
```

Step 4 – Build Docker Image

```
meghana@meghana-HP-Laptop-15s-dr3xxx: /program_devops_2$ docker build -t program_devops_2 .
[+] Building 33.5s (12/12) FINISHED
=> [internal] load build definition from Dockerfile
=> == transferring dockerfile: 320B
=> [internal] load metadata for docker.io/library/node:18-alpine
=> [auth] library/node:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> == transferring context: 2B
=> [builder 1/5] FROM docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8ca09d9e
=> == resolve docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8ca09d9e
=> == sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8ca09d9e 7.67kB / 7.67kB
=> == sha256:929b04d7c782f04f615cf785488fed452b6569f87c73ff66ad553a7554f0096 1.72kB / 1.72kB
=> == sha256:ee77c6cd7c1886ecc802ad0cedf3a8ec1ea27d1fb96162bf03dd3718839b8da 6.18kB / 6.18kB
=> == sha256:f18232174bc91741fdf3da96d85011892101a832a93a308b79e99e69c2d5c878 3.64MB / 3.64MB
=> == sha256:dd71dde834b5c203d162902e6b8994cb2389ae049a0eabc4defea161b2b5a3d0e 48.01MB / 48.01MB
=> == sha256:1e544c89cee5c0826c540ab06d4b6b491c96edab1837f430bd47f8d26782d6e3 1.26MB / 1.26MB
=> == extracting sha256:f18232174bc91741fdf3da96d85011892101a832a93a308b79e99e69c2d5c878
=> == sha256:25ff2da83641988f65c3a74d88409d6b1b62ccfaab220b9ea70b88df5a2e0549 446B / 446B
=> == extracting sha256:dd71dde834b5c203d162902e6b8994cb2389ae049a0eabc4defea161b2b5a3d0e
=> == extracting sha256:1e544c89cee5c0826c540ab06d4b6b491c96edab1837f430bd47f8d26782d6e3
=> == extracting sha256:25ff2da83641988f65c3a74d88409d6b1b62ccfaab220b9ea70b88df5a2e0549
=> [internal] load build context
=> == transferring context: 596B
=> [builder 2/5] WORKDIR /app
=> [builder 3/5] COPY package.json .
=> [builder 4/5] RUN npm install
=> [builder 5/5] COPY server.js .
=> [stage 1 3/3] COPY --from=builder /app .
=> exporting to image
=> == exporting layers
=> == writing image sha256:0cf8958f2b1d52a2526d6d96fea22e46c1035c07d4a5153a19c162aebd3e7b
=> == naming to docker.io/library/program_devops_2
meghana@meghana-HP-Laptop-15s-dr3xxx: /program_devops_2$
```

Step 5 – Run Docker Container

`docker run -p 3000:3000 program_devops_2`

Output Verification

Application URL:

<http://localhost:3000>

